

Lab 3: Architectural Justification

Micro-Lending & Peer Credit Risk Assessor (Problem Statement #33)

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Architecture Selection: We chose Layered Architecture for the Micro-Lending & Peer Credit Risk Assessor System.

Two reasons for this choice:

Reason 1 — Separation of concerns between UI and lending logic

The Presentation layer (Borrower / Lender Portal) is fully decoupled from the Business layer, which houses the Risk Assessment Engine, Loan Matching Service, and Default Monitoring Service. UI changes — new onboarding flows, dashboards, or notifications for borrowers and peer lenders — can be made without touching the heuristic risk-scoring rules or repayment-scheduling logic, and vice versa.

Reason 2 — Centralized, consistent financial data

The platform's NFR-001 requires EMI distribution and repayment transactions to run inside ACID transactions to prevent double-spending. A layered architecture keeps all persistence behind a single Data layer (Transaction & Ledger DB) accessed only through well-defined interfaces, making it far simpler to guarantee consistency than a microservices design where loan, borrower, and ledger data would be scattered across independently-owned datastores.

Security Advantage

All access to the Transaction & Ledger DB is funneled through a small, auditable set of Business-layer interfaces — BorrowerHistory, LoanRecords, and RepaymentLedger. The Presentation layer never talks to the database directly, so the Borrower/Lender Portal (the component most exposed to untrusted input) has no direct path to sensitive financial records, shrinking the attack surface for credential leaks or injection attacks.

Performance Benefit

Because the Risk Assessment Engine, Loan Matching Service, and Default Monitoring Service sit together in one Business layer rather than being split into independently-deployed microservices, calls such as RiskProfile → Loan Matching happen with in-process, low-latency calls instead of network round-trips. This keeps credit-risk-tier computation fast enough to feel immediate to a borrower submitting a loan request, as required by FR-001.

System Components & Interface Specifications

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Components List

1	Borrower & Lender Portal	Presentation	FR-001, FR-002, FR-004 (submit financial profile, request loan match, view live status)
2	Loan Workflow Manager	Business	FR-003, FR-005 (lifecycle state machine: applied → matched → repaying → closed/defaulted; status push to clients)

3	Risk & Matching Engine	Business	FR-001, FR-002 (heuristic risk scoring; auto-match to eligible, risk-compatible lender)
4	Notification & Escalation Service	Business	FR-005 (SMS/email escalation for stalled repayments and default warnings)
5	Auth & Access Control	Business (Crosscutting)	NFR-002 (role-based access for borrowers and lenders)
6	Transaction & Ledger DB	Data	NFR-001 (ACID transactions); persistence for profiles, loans, and repayment history

Interfaces

1	ProfileSubmission	Loan Workflow Manager	Borrower & Lender Portal	Borrower submits financial profile (income, employment, debt) for a new loan request
2	StatusQuery	Loan Workflow Manager	Borrower & Lender Portal	Read-only live loan and repayment status lookups for borrowers and lenders
3	RiskScoreLookup	Risk & Matching Engine	Loan Workflow Manager	Provide the computed risk tier (Low/Med/High) for a submitted borrower profile
4	AssignMatch	Risk & Matching Engine	Loan Workflow Manager	Route a newly risk-scored borrower to a free, risk compatible peer lender
5	AccessCheck	Auth & Access Control	Loan Workflow Manager, Risk & Matching Engine, Notification & Escalation Service	Enforce role-based access before any business-layer operation
6	EscalationTrigger	Notification & Escalation Service	Loan Workflow Manager	Fire SMS/email alert when a repayment misses its due date or risk crosses threshold
7	LedgerAccess	Transaction & Ledger DB	Loan Workflow Manager, Risk & Matching Engine	Persist and retrieve loan records, repayment ledger, and profile/status history

Component Diagram:

